

Research Questions

Primary Question:

Can we use transfer-market and performance data to build a predictive model that estimates a player's expected market value?

Secondary Question:

Using the residuals from this model (actual value – expected value), can we identify which players are over- or under-valued and assess the efficiency of the transfer market

Motivation

Player market value is often treated as an indicator of talent and potential, but transfer fees are influenced by many non-performance factors such as media attention, reputation, or league exposure. As sports analytics students, understanding whether market prices are accurate in reflecting the actual performance of players can help clubs identify which players are valued correctly within the market to help benefit their team's needs.

A player who is overvalued can cause a significant loss in money resulting in financial burden for clubs in soccer. This can cause detrimental effects to team performance, especially for high contract players with long contract lengths who are not consistently performing to their expected level based on the market value they were assigned.

Previous work from StatsBomb, FBref, and academic studies has analyzed market value or performance metrics independently, but relatively few examine how well the market prices players relative to their actual performance.

Our project extends this gap by applying the idea of an “expected market value”, which is similar to expected metrics like xG or xwOBA, to evaluate pricing fairness. By creating a model to predict market value, we can understand the underlying reasons for placing players within a specific market value while also assessing the over or undervaluation of players within these brackets. This hopefully will help us answer our two research questions.

Objective

Our objective is to estimate expected market value using machine learning models and use the resulting residuals as a measure of whether pricing is accurate or not within the transfer market.

By comparing actual market prices to the model's expected market prices, we can conclude

1. Whether the transfer market systematically over or under values certain players
2. Whether valuations align with the actual performance of players according to data

3. Whether any patterns or biases (e.g., position, nationality, age) appear in the pricing of these players

Through this approach, we provide a framework that connects predictive modeling with an extremely important part of a soccer club's success, their understanding of the transfer market.

Data Collection and Preparation

Data Sources:

We used two distinct public soccer datasets:

1. FBref (<https://fbref.com/en/>) – provides official league performance data across various players in different leagues.
2. Transfermarkt (<https://transfermarkt.com/>) – provides player market values, transfer fees, contract information, club history, and positional data.

To collect these datasets, we used the worldfootballR R package (<https://github.com/JaseZiv/worldfootballR>) as a data-access tool to download FBRef and web scrape Transfermarkt statistics. To aid in analysis, we kept three final datasets which contained just data from FBRef, just data from Transfermarkt and a combined dataset joined based on player name and the year they started. We also combined players who played multiple positions across the four main position categories: goalkeeper, defense, midfield, and attack. During analysis, we'd omit NA values, but did not remove them directly from the dataset as removing rows with NA values could significantly affect the accuracy of the results we obtained.

Unit of Observation and Time Frame:

The unit of observation is player-season across Europe's top five leagues (Premier League, La Liga, Bundesliga, Serie A, Ligue 1). The dataset covers the 2014–2024 seasons (10 years). The combined dataset contains approximately 40,000 player-season observations.

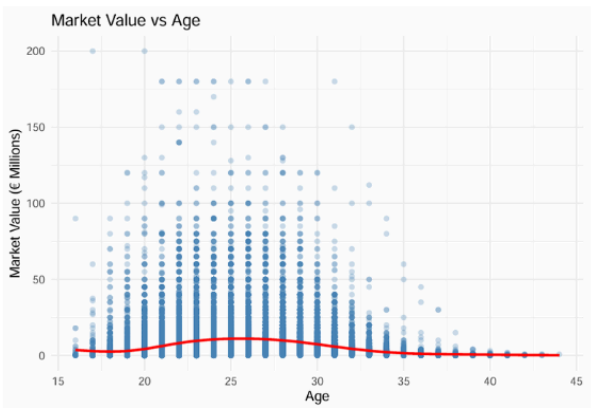
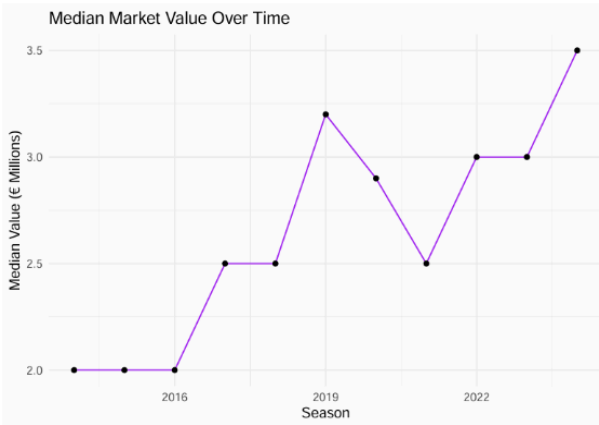
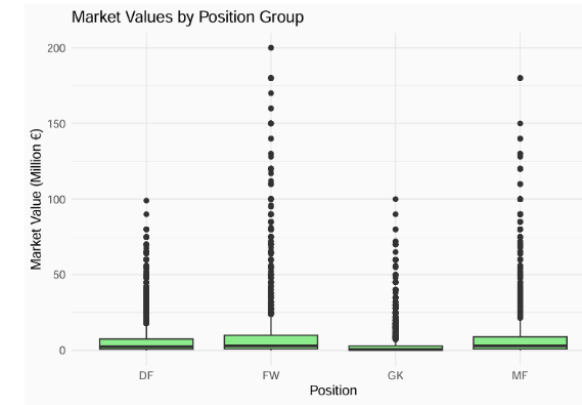
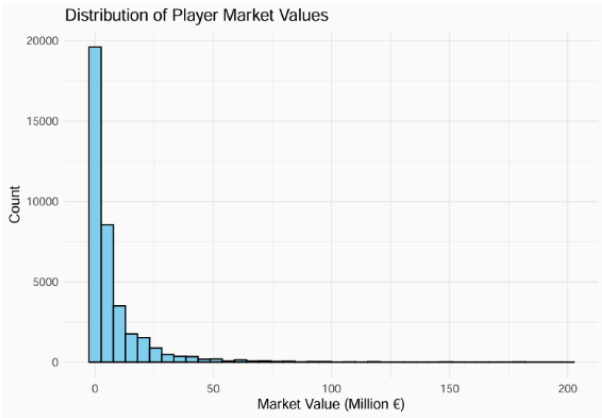
Key Variables:

- **FBRef:** Goals, Assists, xG, Passes, Shot Creation, Defensive Actions
- **Transfermarkt:** Market Value, Position, Age, League

These variables jointly allow us to examine the relationship between measurable on-field performance and market valuation, train a model predicting expected market value, and analyze residuals to identify under- and over-valued players as indicators of pricing efficiency.

Exploratory Data Analysis

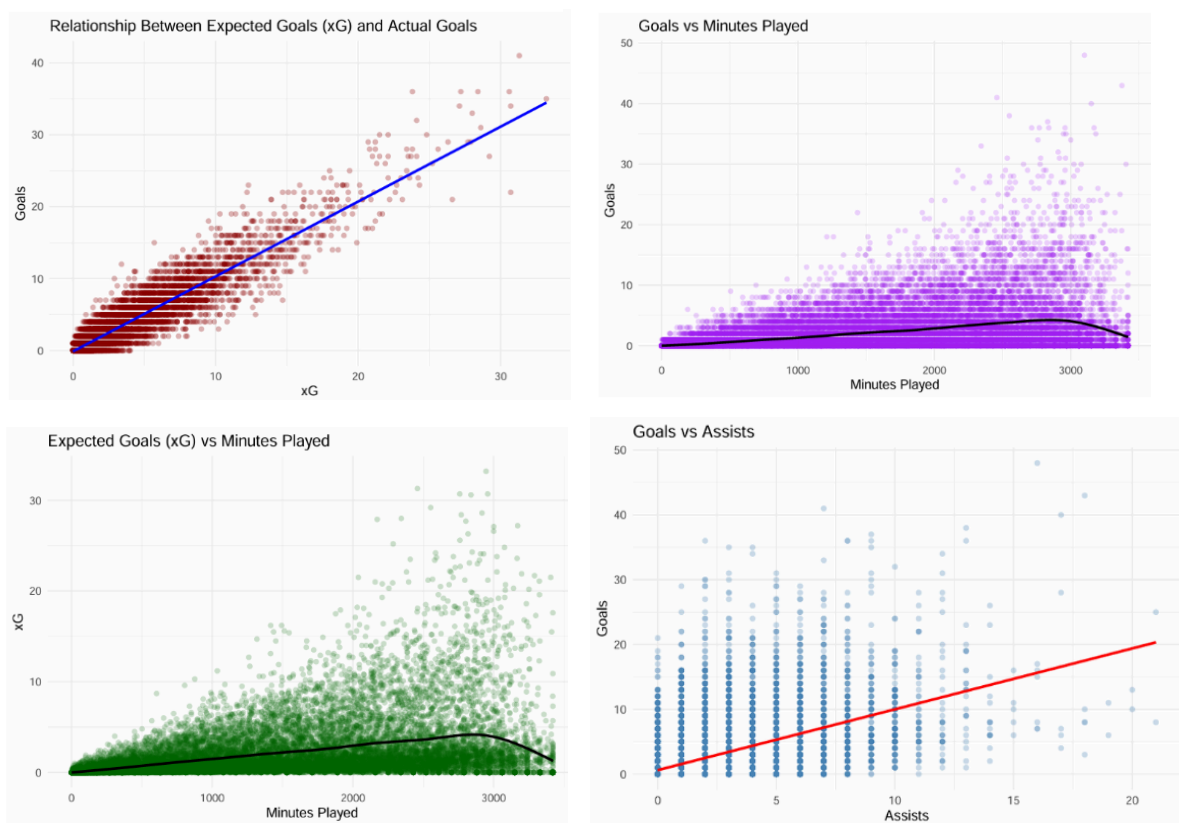
Transfermarkt:



Graph	Statistics
Distribution of Player Market Values	Median: 2.5 M€ Max: 200 M€ IQR: 7.2 M€
Market Values vs Position Group	<u>Median MVs:</u> Defenders: 2.5M€ Forwards: 3M€ Goalkeepers: 0.8M€ Midfielders: 3M€
Market Value Over Time	Correlation: -0.011
Market Value vs Age	Correlation: 0.87

Above are the key findings from the EDA for the Transfermarkt dataset. The key variables were obviously the outcome of market value and comparing them with year, age, and position group. As can be seen, there is a significant increase in the median market values of players over time (not adjusting for inflation which could also have an effect on findings). Secondly, although there is no significant correlation between age (treated as a numeric variable) and market value, the spike occurring at the mid-20s point shows significance with the market values increasing for players around that age. Regarding the market value spread, the market values are seen to be very rightly skewed, but spread out nonetheless. Finally, it is seen that defensive positions have smaller market values on average than midfielders and forwards, with there also being a larger amount of high outliers for market values within the more attacking positions.

FBRef:



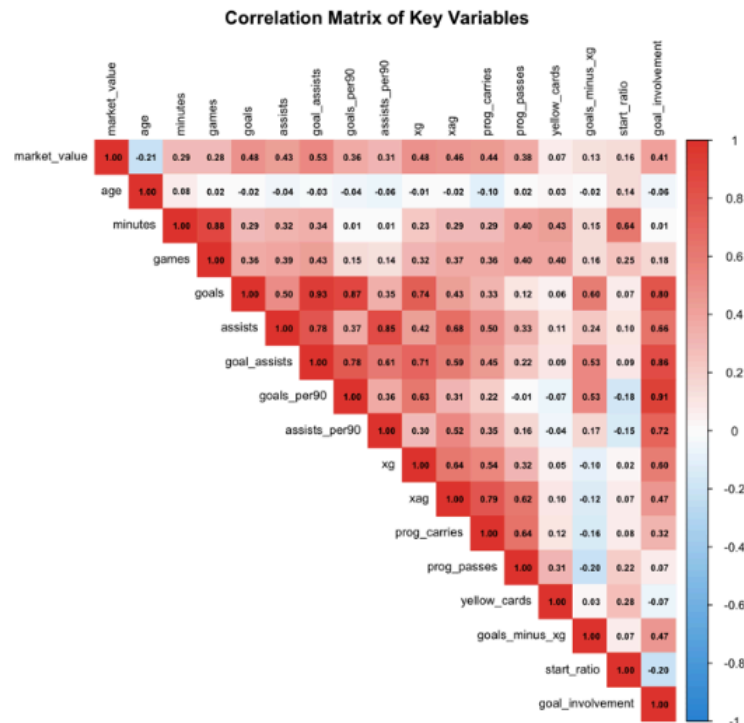
<u>Graph</u>	<u>Statistics</u>
Expected Goals vs Actual Goals	Correlation: 0.94
Goals vs Minutes Played	Correlation: 0.40

Expected Goals vs Minutes Played	Correlation: 0.42
Goals vs Assists	Correlation: 0.56

Above are the key findings from the FBRef dataset. The key variables analyzed here are goals, assists, expected goals, and minutes played. Goals being a primary variable within the analysis due to its strong correlation with assists, expected goals, and minutes played. Expected goals also have a strong correlation with minutes played.

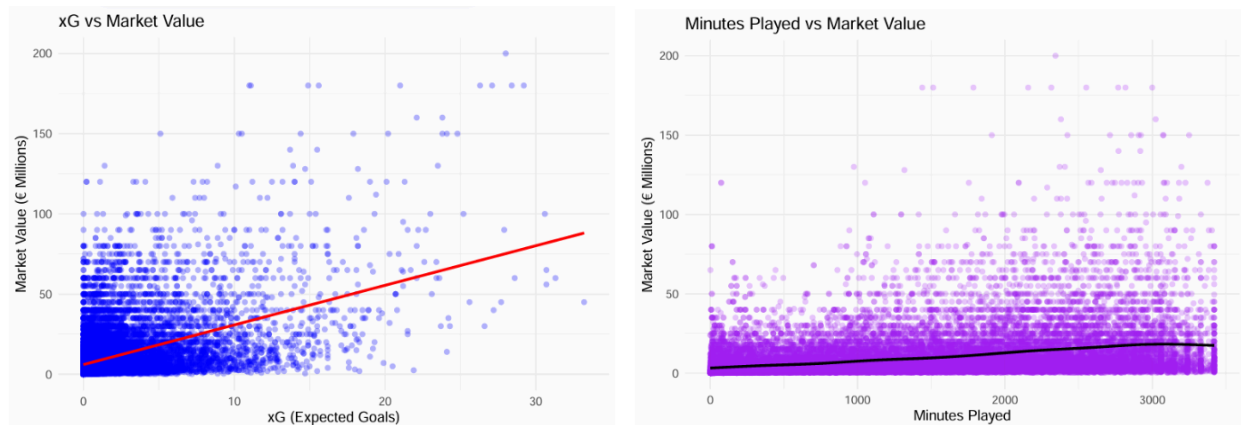
The findings from this EDA indicate specifically the validity of the expected goals metric in having it present in the model. One can argue one of two things, the metric will cause multicollinearity when training machine learning models on it, or it will be a strong predictor. During EDA, we weren't sure whether to remove or keep expected goals. What will be seen later is the use of tree models as a major part of our analysis, tree-based models are known to handle multicollinearity quite well. To maintain consistency across the analysis, we decided to keep the variable due to it still being a possibly valuable metric in predicting market value and checking for how over or under valued a player is.

Combined FBRef and Transfrmarket:



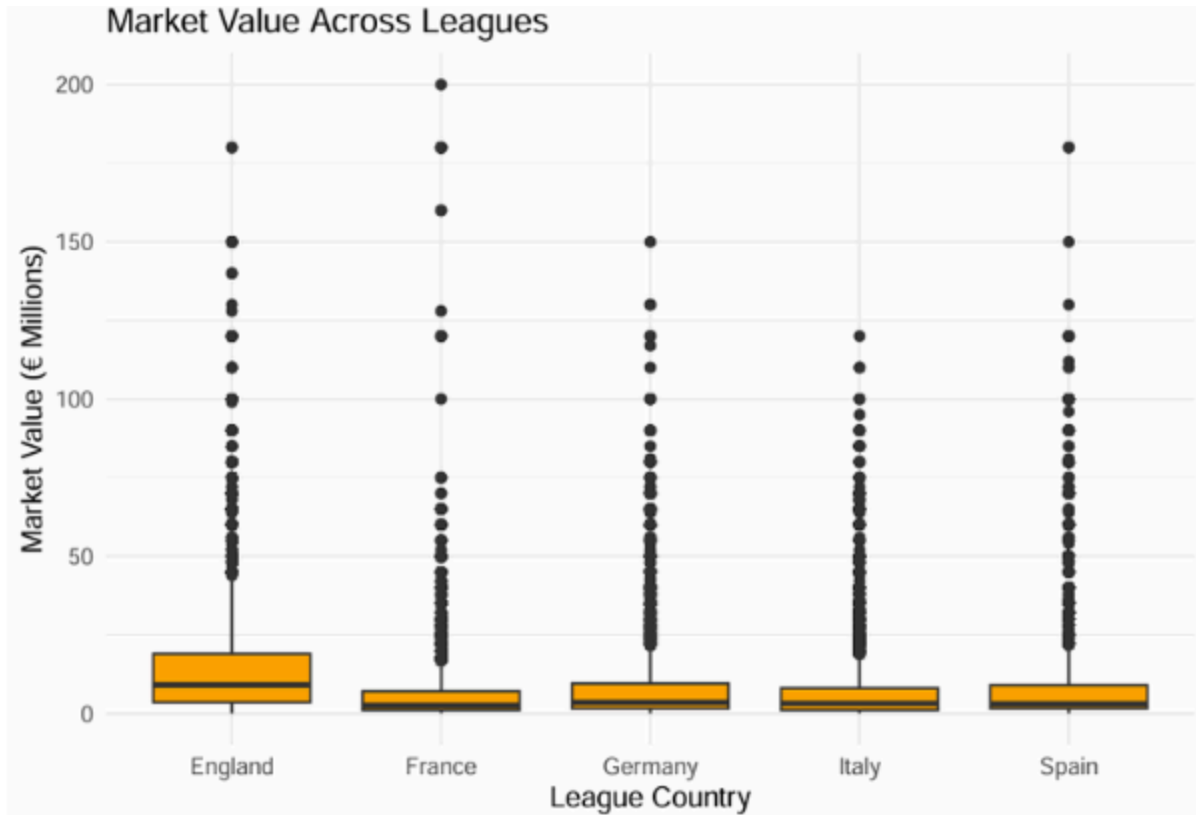
To start off, this correlation matrix enabled us to study the overall correlation between specific variables in the combined dataset and accordingly visualize some of the key highly correlation variables. The top row of the matrix is of highest importance due to it showcasing the correlation between variables and market value.

Expected goals were strongly correlated with a lot of other variables having a strong correlation with market value (goals, assists, goals per 90, goals + assists are some examples). Hence, to view the high-level relationship between these variables and market value along with the nature of data, we decided to measure the xG vs market value relationship visually. Another relationship we were interested in was the minutes played and market value.



The calculated correlations using the matrix was 0.48 for expected goals and 0.29 for minutes played. Oddly enough though when graphing, the correlations turned out to be 0.47 and 0.33 for expected goals and minutes played respectfully. There was very probably a difference in the way the correlation matrix function and the correlation itself handled missing values. We think it is important to report this inconsistency just to be honest about the findings. In our opinion, it is not significant enough to make a difference for the analysis as they produce the same results, but it is still interesting to note the same.

Anyway, the variables here have a fairly strong correlation with market value, obviously not extremely strong, but strong enough to be relevant in predicting market value. Finally, we were curious about how the league of players affect their market value, as this is not reported in the correlation matrix given the league is categorical.



To report the median market values, the Premier League (England) had a median market value of 9 million euros, compared to the rest of the leagues having their median market values ranging between 2.5-3.5 million euros. This shows a significant difference in the demand for Premier League players. This can be backed by the fact that the Premier League is considered the most competitive league, so the players would effectively be highly valued.

Now to put all these findings to fruition, we will now move towards the actual analysis used to answer our research questions.

Methodology

The dataset was split into training (80%) and testing (20%) subsets. We evaluated four supervised models:

- **Linear Regression**, a baseline model assuming additive linear relationships.
- **Elastic Net Regression**, combining L1/L2 regularization to address multicollinearity
- **Random Forest**, a nonlinear ensemble method robust to outliers and correlated features.
- **XGBoost**, a gradient boosting model that iteratively corrects prediction errors and excels at capturing interactions.

Models were evaluated using RMSE (log-scale and raw-scale) and R^2 . The best-performing model was selected for residual-based market efficiency analysis.

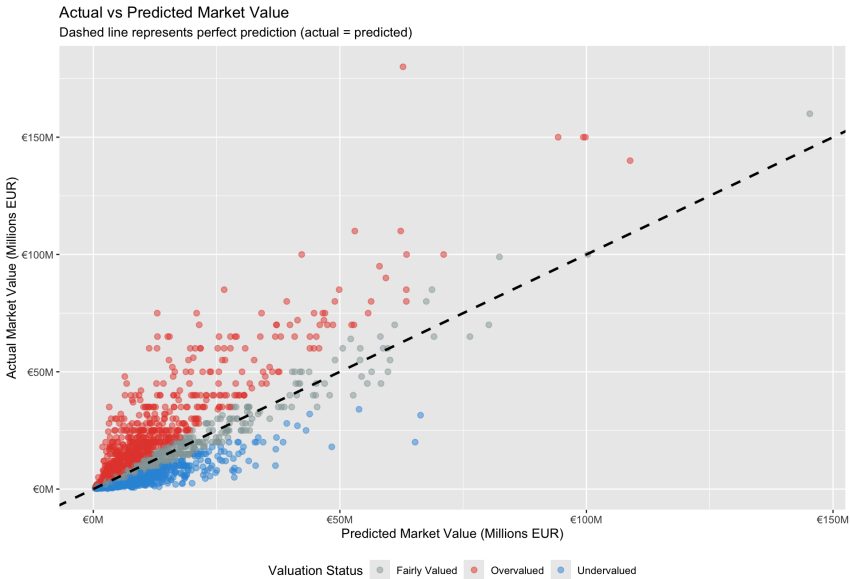
Results

Model comparison revealed clear performance differences across the four predictive approaches. The linear models like Linear Regression and Elastic Net were able to capture broad valuation patterns but struggled to represent the nonlinear relationships in the data. In contrast, tree-based models demonstrated substantially stronger predictive capability. The best overall performance came from XGBoost, which achieved an RMSE (Log) of 0.6029, RMSE of €10.2381M, and an R^2 of 0.6311 which was the highest among all models. These results reflect XGBoost’s ability to iteratively correct residual errors, model complex interactions, and handle nonlinear structure, making it the most effective model for predicting player market values in this project.

Table 5: Model Performance Comparison

Model	RMSE (Log)	RMSE (€M)	R-Squared
XGBoost	0.6029	10.2381	0.6311
Random Forest	0.6207	11.3049	0.6152
Elastic Net	0.6527	11.1186	0.5671
Linear Regression	0.6535	11.1477	0.5661

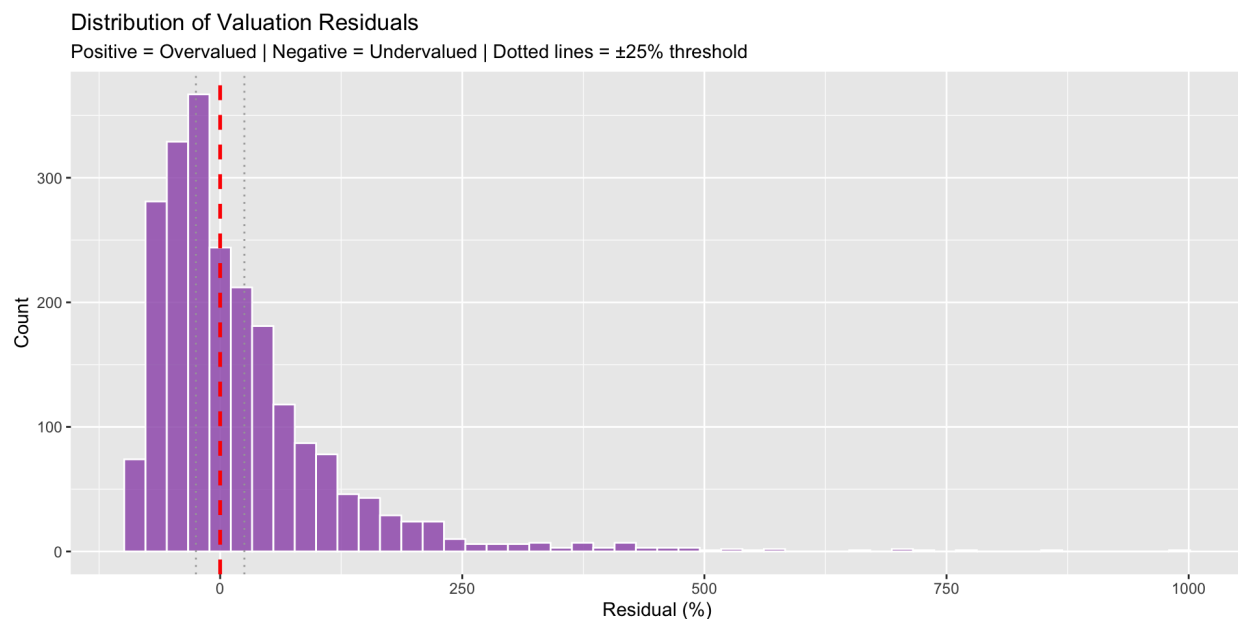
To evaluate model calibration, here is a scatterplot of actual versus predicted market values under the XGBoost model. Points lying above the dotted line indicate players whose market values exceed model expectations (overvalued), while points below correspond to undervalued players. The clear dispersion around the diagonal indicates that although performance metrics explain substantial variation in values, the transfer market has significant noise.



These results align with earlier EDA findings. For example, forwards exhibited higher variance in value. And because XGBoost demonstrated consistently lower RMSE and higher R^2 than all other models, it was selected for full residual analysis.

Discussion

The results of this study demonstrate that while the XGBoost model captured a substantial portion of the structure behind player market valuations, significant deviations remain between predicted and actual values. The Distribution of Valuation Residuals reveals a wide spread of positive and negative errors, with many players falling outside the $\pm 25\%$ band typically considered reasonable. This distribution illustrates that the transfer market does not strictly reflect observable performance; instead, it incorporates non-performance factors such as reputation, media presence, contract situations, and league visibility.



Further insights emerge in the lists of Top Overvalued Players and Top Undervalued Players. Overvalued players tend to be those with established reputational capital like international experience, past peak performance, or high-profile club affiliations which causes their market values to exceed what their current statistical contributions justify. On the other hand, undervalued players often come from smaller clubs or lower-visibility leagues, despite strong statistical performance. This suggests opportunities for analytically inclined clubs to identify undervalued talent that the wider market fails to price efficiently.

Table 7: Top 15 Most Overvalued Players

Player	Age	Position	League	Actual (€M)	Predicted (€M)	Residual (%)
Julian Draxler	23	FWD	Ligue 1	35.0	3.23	983.9
Georginio Wijnaldum	31	MID	Ligue 1	18.0	1.87	861.0
David Luiz	28	DEF	Ligue 1	28.0	3.18	780.3
Milan Škriniar	29	DEF	Ligue 1	30.0	3.63	725.5
Boubakary Soumaré	21	DEF	Ligue 1	24.5	3.04	706.0
Arjen Robben	32	FWD	Bundesliga	20.0	2.51	697.8
Kalidou Koulibaly	30	DEF	Serie A	48.0	6.39	651.4
André Schürrle	24	FWD	Bundesliga	30.0	4.46	573.2
Manuel Neuer	30	GK	Bundesliga	45.0	6.76	565.6
Reinildo Mandava	28	DEF	Ligue 1	22.0	3.40	547.8
Blaise Matuidi	30	MID	Ligue 1	30.0	4.73	534.8
Marco Verratti	22	MID	Ligue 1	40.0	6.42	523.3
Sven Bender	26	DEF	Bundesliga	16.0	2.65	504.6
Thilo Kehrer	24	DEF	Ligue 1	25.0	4.30	481.7
Marquinhos	27	DEF	Ligue 1	75.0	12.96	478.6

Table 8: Top 15 Most Undervalued Players

Player	Age	Position	League	Actual (€M)	Predicted (€M)	Residual (%)
Ousmane Camara	21	FWD	Ligue 1	0.15	2.85	-94.7
Panagiotis Tachtsidis	29	DEF	Serie A	0.95	13.35	-92.9
Fabio Lucioni	32	DEF	Serie A	0.72	8.35	-91.3
Lassine Sinayoko	23	FWD	Ligue 1	1.20	12.41	-90.3
Antonio Caracciolo	28	DEF	Serie A	0.80	7.94	-89.9
Eros Pisano	29	DEF	Serie A	0.80	7.90	-89.9
Alban Lafont	17	GK	Ligue 1	0.80	7.78	-89.7
Simone Calvano	24	MID	Serie A	0.40	3.81	-89.5
Nehuén Paz	27	DEF	Serie A	0.32	3.09	-89.5
Marco Höger	28	DEF	Bundesliga	2.00	16.69	-88.0
Eros Pisano	28	DEF	Serie A	0.50	4.01	-87.5
Tom Heaton	29	GK	Premier League	2.50	20.03	-87.5
Mattia Valoti	24	MID	Serie A	1.00	7.72	-87.1
Iván Balliu	26	DEF	Ligue 1	1.00	7.57	-86.8
Riccardo Saponara	28	MID	Serie A	2.40	18.05	-86.7

Defenders show positive residuals on average, indicating that they are often valued above their predicted model output. While goalkeepers display strong negative residuals, implying underpricing. These findings are shown in the Market Efficiency by Position table. This pattern reflects long-standing challenges in quantifying defensive contributions, particularly those not captured by traditional event-based metrics.

Table 9: Market Efficiency by Position

Position	Count	Mean Residual (%)	Median Residual (%)	SD Residual (%)	% Overvalued	% Undervalued
DEF	998	27.27	-2.27	106.76	36.47	35.77
FWD	613	18.12	-6.90	92.61	35.89	35.73
GK	146	4.70	-21.52	90.61	30.82	46.58
MID	457	18.38	-14.64	103.18	32.82	40.04

Similarly, the Market Efficiency by League table highlights variation in pricing norms across leagues. Top European leagues (e.g., Premier League, Bundesliga) display higher average positive residuals, suggesting inflated valuations relative to performance. In contrast, less globally visible leagues show more negative residuals, indicating undervaluation. These discrepancies likely stem from broadcast visibility, commercial revenue, league reputation, and differing expectations of player development and competitive level.

Table 10: Market Efficiency by League

League	Count	Mean Residual (%)	Median Residual (%)	% Overvalued	% Undervalued
Bundesliga	494	13.19	-15.93	28.54	41.70
Premier League	558	19.43	8.48	40.68	29.39
Serie A	653	24.15	-3.88	36.91	37.21
Ligue 1	509	28.06	-11.41	33.40	42.04

Conclusion

This study used machine-learning models to examine the extent to which on-field performance metrics predict player market values and whether deviations from predicted values reveal inefficiencies in the transfer market. The findings offer three key contributions.

First, performance metrics are strong predictors of market valuation. XGBoost's R^2 of 0.6311 indicates that modeling quantitative performance data can reconstruct much of the valuation structure observed in the transfer market. Second, substantial mispricing remains. Both over- and undervaluation patterns appear consistently across players, positions, and leagues. These deviations highlight the influence of subjective factors such as reputation, tactical role visibility, league prestige, agent negotiation power, and media narratives. Third, position and league biases suggest that clubs could exploit undervalued player markets by strategically targeting goalkeepers, defenders, or players in lower-visibility leagues who consistently outperform their market price.

Future research could extend this work by integrating additional data sources such as wage information, contract length, injury history, etc. or analyzing leagues outside of Europe to improve the predictive power of valuation models.